

Impacts to Productivity in the Painting Trades

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Abstract—The “Painting Trades” include commercial and industrial painters and drywall finishers, which are normally represented by the same labor union for collective bargaining purposes. They are also members of the same construction trade organization, the Finishing Contractors Association (FCA). Labor is the greatest variable cost of performance for these trades and so, labor productivity is a critical concern of these trade contractors. New and evolving technologies for the collecting of labor productivity data have emerged in recent years, however this data cannot be used in absence of some evidence exogenous environmental influences actually impact the trade contractors’ productivity. The FCA funded this research to better understand the factors which impact labor productivity and to what degree. The FCA collected data on 140 commercial painting projects, 56 industrial painting projects, and 147 drywall finishing projects. Fifteen productivity variables were derived from a review of the literature and in discussions with an expert panel of industry executives, selected by the FCA Executive Director¹. The data was collected by the FCA using an online survey sent to FCA member contractors. Correlation matrices and single linear regression analyses were performed on the data which revealed the majority of the variables were negatively correlated with labor productivity and were statistically significant at very high levels. This is really project and construction management research, however even though the technology now exists to collect production data in the field, without underlying studies, like this one, linking changes in productivity to exogenous impacts, merely collecting data will be unhelpful.

I. INTRODUCTION

Construction is one of the largest sectors of the U.S. economy, directly employing five percent of all U.S. workers. Construction costs are broken into the following categories: Labor, Materials, Equipment, Supervision, Overhead and Profit. Of these, the most highly variable cost is labor. When construction contractors estimate their costs they first do a quantity take-off of the work to be completed, in the painting trades, this usually means quantifying the square feet of area to be painted or finished. They then apply a labor productivity rate, typically in square feet per man hour, to calculate the total number of hours of labor required. The productivity rate is usually derived by each contractor based on its own prior work experience (as opposed to some common standard) [1]. Therefore, labor productivity is critically important to a contractor’s cost performance on a project and its overall profitability as a company. With new and emerging technologies used to collect production data, such as electronic

tablets linked to Computer Aided Design or CAD documents, understanding the exogenous factors which impact the contractor’s productivity and to what degree, aids trade contractors when seeing an equitable adjustment² to their contract price from a general contractor or owner for those impacts. The high cost and fast paced development of technology investments, such as chip fabrication facilities, or “Fabs” make understanding these even more important for the tech industry.

The field of labor productivity investigations in the construction sector is relatively mature. The first trade productivity impact studies were performed by the National Electrical Contractors Association (NECA) in the late 1960’s and early 1970’s [2, 3], followed by the first publication of the Mechanical Contractors Association of America’s (MCAA) *Change Orders, Productivity, Overtime* in the 1970’s, which has since been reissued and updated a number of times since, including [4]. In 1979, the U.S. Army Corps of Engineers published its *Modification Impact Guide* [5], or Corps civil projects, which has since been taken out of circulation, however the Corps has never repudiated its findings and conclusions [6]. Of the roughly 2,300 papers published citations in our database, fewer than 150 have been published since 2010, including one recently published by two of the authors of this paper, involving the metal stud framing and gypsum drywall trades [7]. It was that research which prompted the FCA to fund similar research in the painting trades, which had not previously been studied.

In the construction industry, labor productivity is a function of three functional variables: the environment in which the is to be executed; the manner in which the contractor is allowed to execute the work; and, the internal capabilities of the construction firm. Within these functional variables there are several specific factors commonly regarded as having a significant potential impact on labor productivity, and thus the cost of the work performed [8] [9] [10] These factors have been studied extensively [11] [12] and can be broken into two broad groups: controllable, those factors associated with the contractor’s internal operations, its estimate, and project team; and, uncontrollable, those factors

¹ See Appendix A for members.

² The term “Equitable Adjustment” is something of a term of art in the construction trades and the law. When a contractor feels it has been damaged, that its cost of production has been increased due to factors beyond its control, it will first seek a contractual remedy, a change order to the contract price, but if the change order is denied, the contractor will file a claim for an equitable remedy or equitable adjustment.

which are outside of the trade contractor's control, such as extreme weather, changes to the work, schedule disruptions, and other items controlled by the general contractor or the owner.

Ultimately, the question becomes, just because the MCAA or NECA found factors impacted the productivity of their trades, why would other trades seem to be able to piggy-back on that research to assert their own claims? Oglesby's 1989 book [13] adds fuel to that argument, because he showed there were different levels of productivity impact based on trade. Masons, for example, were harder impacted by extremely cold weather than other trades (based on earlier work by [14]). The Northwest Wall & Ceiling Bureau (NWCBC) undertook their own study of labor productivity impacts precisely because its own trade contractors were routinely criticized for relying on studies like the Corps of Engineers, MCAA and NECA, for their Requests for Equitable Adjustments [15]. The NWCBC's study also included taping and finishing of gypsum drywall, but we believe this study is actually superior to the earlier study, because we learned from the earlier study, we needed to give context to the rankings of impact. In the NWCBC study, respondents were asked to rank the impact from zero to four, with zero being no impact and four being severe impact. But we did not give the respondents any context or narrative of what a four really meant. (See Appendix B for the actual data collection instrument.)

As in the NWCBC study, we found many or most of the impact variables were highly (positively) correlated, which leads to the conclusion, as in the earlier study, when projects "go bad" they go bad on multiple levels, which are highly correlated. This has been referred to as the "ripple effect" [16-20], where the first domino to fall may be one issue, but it impacts many issues down the line, and in fact, perhaps all the dominos fall.

For this study, we used the commonly accepted definition of productivity loss: "the decline of labor efficiency due to specific causes from the level which could have been expected except for the cause(s) under examination" [21].

II. LITERATURE REVIEW

As noted in the INTRODUCTION, there have been a number of studies on labor productivity in the construction trades by major trade organizations such as NECA, MCAA, the NWCBC [22] [4] [10], the government [5, 23], business groups [24-26] and other professional organizations [11, 27-33] and academic institutions [34-47]. In 1990, Thomas [12], published a comprehensive review of the published works on labor productivity in the construction trades, which has been recently updated by these authors for a text book currently being used at the University of California at Berkeley³. The book includes a database of more than 2,200 citations to published works on the topic of construction labor productivity, scheduling, and delay impacts in the various

trades, though some of the references are also to legal holdings from the various courts in the land, including state and federal courts, and the courts of contract appeals. (There are various other rulings, such as those by the General Services Administration, which bear on construction claims, which are not included in the database.)

In addition to being one of the largest sectors of the economy, construction for hire is also one of the oldest trades⁴, predating our own republic by centuries. As such, the actions or activities which impact productive use of labor on a construction job have been well known for a very long time, and extensively studied. There have been no new impact items identified, which negatively impact construction labor productivity in at least thirty years and more likely longer. The recent research has been driven by applications in the various construction trades to determine the relative impact of the same or similar condition on a previously unstudied trade, more so than the discovery of new impacts (though with new computer systems, specifically Building Information Modeling (BIM) software, researchers may soon start considering a BIM impact). In 1989 Oglesby [13] published a compilation of different trades productivity impacts due to temperature and humidity (bricklayers suffered much greater productivity losses due to cold weather than equipment operators). Each trade, from earthwork [48-50] and concrete operations [51-58] to mechanical [16, 41, 59-63] and electrical contractors [2, 3, 16, 22, 64-68]. However, prior to this research, no papers have been published on the impacts to labor productivity in the commercial and industrial painting trades and only one in the taping and finishing of gypsum drywall trade [7] (by two of the authors of this research).

The most commonly studied impacts to productivity in the construction trades include: Scheduling and Acceleration which result in: congestion, overtime, dilution of supervision, disruption and fragmentation, out-of-sequence work, ramp-up/ramp-down of the workforce; Changes in the work; and, Extreme Weather. Other variables considered include: quality of plans and specifications, overzealous inspections, administration of the work (by the general contractor or owner), workforce and problems with vendor deliveries.

A. SCHEDLING & ACCELERATION

The functions of management are widely thought to be: Planning, Staffing, Measuring and Correcting, and there are few things on a construction project which can be measured and corrected other than Schedule, Cost, and Quality Performance. While all three activities are tightly connected, delays in the performance of the work have a greater impact on cost than just about anything else, and has been the subject of more than 200 published works, some dating back to the 1960's while others are much more recent [26, 33, 37, 40, 46, 69-89].

³ The book has been copyrighted but is currently only available to students at UC Berkeley electronically.

⁴ From the Code of Hammurabi around 1780 BCE: Law #228: "If a builder build a house for someone and complete it, he shall give him a fee of two shekels in money for each sar of surface."

When a project falls behind schedule and the owner is unwilling to increase the performance period⁵ and threatens to assess actual or liquidated damages, the general contractor often has little choice but to accelerate the remaining work. But acceleration has its own downside in the form of congestion and other problems noted above, the impacts of which are discussed in brief below.

1. Congestion

Congestion occurs when a trade has too many people or too much material in front of them so they can't get their work done efficiently. This usually comes about as a result of the stacking of trades (scheduling multiple trades in an area at the same time) crowding or overmanning (demanding a trade to "crew-up" to finish faster), and construction material storage in work areas (where other trades are scheduled to work). [90]

A simple example might be if you have a room with four walls which would take a painter four hours to finish, one hour per wall. But instead, the general contractor demands the painter to put eight painters in the room and get it done in a half-hour. The likelihood is the eight workers would be falling all over themselves and instead of taking four man-hours labor, it takes them 8 man-hours to paint the room. At the same time, the general contractor decides to schedule its acoustical ceiling subcontractor to install the ceiling tiles, since the ceiling isn't the walls. All the while, the room was being used as a storage location by the mechanical and electrical subcontractors.

2. Overtime

Another common acceleration tool is to have workers work overtime on the belief that more hours worked equals more production. Of course, this assumes productivity remains constant, during the overtime period, and for some time, research shows this is a valid assumption. However, research also shows when overtime continues for extended periods, productivity drops, and after enough time, the marginal gains from working more hours is lost. [26, 68, 91, 92] Another way to inject more hours into the workweek is to schedule multiple shifts. Shift work can be inherently unproductive in some cases due to labor union rules, working in darkness by lights, and fatigue. [26, 68, 91, 92] Alternatively, in very hot climate areas of the country, productivity during cooler night and early morning shifts can make up for these losses.

3. Dilution of Supervision

There is a limit to the number of field labor one person can effectively manage. When, a trade contractor must accelerate

its work by adding labor, its supervisory staff might not be able to keep up with the new personnel; with some workers becoming directionless and unproductive. [12] Dilution of supervision is closely related to overmanning a job (discussed briefly above) on a large scale. In many cases, a trade contractor may not be able to assign existing or hire supervisors to manage the new added workforce, and newly hired supervisors who don't have any prior experience with the contractor may not be productivity themselves. [93]

4. Disruption and Fragmentation

Disruption and subsequent fragmentation of the work is distinctly different from congestion, but have been found to be highly correlated in some trades [10]. The research is not clear as to why fragmentation and crowding are highly correlated, but it may be that a disruption resulting in fragmented work causes a project to fall behind schedule and the only way to get back on schedule is to accelerate the work through overmanning and trade stacking. Work fragmentation is characterized by several conditions including: performing work out of normal or planned sequence; the inability to finish work at one location; and, being forced to move or jump around the project site. [94] Not being able to finish work and having to "go-back" and "pick-up" a minor amount of work is a common complaint of many trade contractors. [10] An example of why go-backs are unproductive consider a drywall contractor who completes a wall to the exterior of a building but must leave out the last three or four feet of the wall at the exterior because exterior window-wall system has not been installed. The finished portion of the wall has mechanical and electrical rough-in installed, insulation and gypsum wallboard, taping, finishing and even painting completed. Once the exterior window system is installed each of the trades must remobilize to the location, and complete the last three or four feet of wall. Owners and general contractors often argue no work was added, there is still the same lineal feet of framing, square feet of insulation, wallboard, and painting, so there shouldn't be any additional cost, or at most simply the added cost to remobilize to back to the work area. But, doing four feet of wall, has a higher unit cost compared with doing one hundred feet of wall. Other ways of costing the impact of disruption and fragmentation include: measured mile analysis [95-98], evaluation of lost learning using learning curves, [99-103] and production curves using production functions. [104]

B. CHANGES IN THE WORK

Construction projects are almost never completed without some change to the scope of work. [105] Excessive changes often resulting from poor quality plans and specifications or major unforeseen conditions, have been shown to impact labor productivity. [4, 16, 17, 41, 106-109] Furthermore, when a change is executed has also been shown to be a separate impact. [110] Possibly the most influential study of the cumulative impact of changes was published by Charles Leonard, in fulfillment of his Master's degree at Concordia College [111], which has been cited in construction for decades, and has been updated from time to time by other

⁵ An owner typically will sign a contract which defines a performance period after which the contractor will be forced to pay the owner either actual or liquidated damages. Actual damages for loss of use are often hard to quantify and therefore many (most even) commercial and industrial construction contracts will "liquidate" their actual damages to a fixed cost per day in the hundreds or thousands of dollars, even more on very large project like a microprocessor chip plant. The threat of being assessed liquidated damages often causes a general contractor to attempt to accelerate the remaining work to finish as close as possible to the original completion date, and minimize its damages.

researchers. [112, 113] Another issue related to cumulative impacts of changes is the “ripple effect” [20], where, like the ripples on a placid pond once it is disturbed by a stone thrown into it, the impacts travel out and disturb all downstream work activities.

C. WEATHER

Weather has several components which impact labor productivity differently for each trade. Weather is comprised of: temperature, rainfall, (non-rainfall) humidity, wind speed, and snow and ice. Trade contractors whose workers are exposed to extremely cold weather can suffer very high productivity losses [14, 23], and an increase in workplace accidents [114]. The presence of snow on the ground can impact activities such as layout of concrete forms and erection of structural steel [115]. Similarly, heavy rainfall can impact earthwork and highway operations [114].

III. ANALYSES

A. PROJECT FORMATION AND DATA COLLECTION

The Finishing Contractors Association (FCA) funded this research in order to better understand the variables which impact the productivity of their members. The FCA is a professional trade organization which represents trade contractors in the painting, finishing, glazing, wall and floor covering trades. In 2009 following the Northwest Walls and Ceilings Association publication [10], on impacts to productivity in its trade group, the FCA funded its own study, using the same research group. The FCA’s executive director formed a group of FCA members and industry professionals titled the, “Labor Productivity Study Work Group consisting of thirteen members (listed in Appendix A) from around the United States, to assist the researchers in identifying variables to be tested and to “cheerlead” the collection of data⁶.

The Work Group met formally and informally over the course of six months, communicating by email and teleconference (since they were spread out from New York to California). The Work Group members represented commercial and industrial painting contractors, finishing contractors, flooring, and glazing contractors⁷. The Work Group settled originally generated several dozen possible variables, but ultimately settled on fifteen variables within three categories of project control: Owner Controlled Issues; Subcontractor Controlled Issues; and General Contractor Controlled, which would be ranked from 1 to 5, where 1 meant no impact and 5 meant the impact was severe. Three other schedule related questions were to be answered simply: yes, or no⁸.

⁶ Construction contractors, as a group, are not the most “open minded” ready to share information about their project data sorts of businessmen. Most fear others will obtain an advantage

⁷ Originally, the study was to include glazing and flooring trades as well as finishing, commercial and industrial painting, however only a handful of glazing project surveys were returned, and no flooring surveys were collected.

⁸ As it turned out, these questions were not found to be significant and were dropped from the analyses.

FCA member contractors were asked to record data from past projects on an online data collection instrument, shown in Appendix B, on their various projects. Members recorded data from 147 taping and finishing projects, 140 commercial and 57 industrial painting projects. The researchers concede this was not a “random” sample, but rather a large, by contemporary construction industry research standards, “convenient” sample, though it should be noted, contractors were asked to provide a range of projects, from good jobs to real catastrophes. The majority of studies in the construction industry labor productivity field studied a handful of projects, many fewer than a dozen⁹, with notable exceptions [10, 100, 112].

Respondents were asked to provide project related information, such as, estimated weeks of work and actual weeks on site; original contract amount and final amount paid; type of work; and, estimated and actual productivity achieved on the job based on objective measures of data collected in the field. Then they were asked to rate the variables on a scale of 1 to 5, discussed above. To assist the respondents the Work Group defined the variable and what each of the rankings meant. For example, the first variable:

Construction Documents: Rate to overall quality of the plans and specifications relative to what you think is required to execute your scope of work.

1. Standard of the Industry. Requires minimal requests for information from the designer.
2. Could not be built without some direction from the designer.
3. Required weekly direction to execute your scope of work.
4. Almost daily direction required.
5. Plans are almost useless. Final project different from the design documents.

By defining the variable and what the rankings were intended to describe, the research team expected to improve on some previous research ([7]) where the meaning of the rankings was not explicitly defined. The problem with leaving the rankings simply: “No Impact” to “Severe Impact”, is that many of the respondents ranked variables as severe, which probably not, given the lack of actual financial impact on project. This contributed to projects becoming statistical clear outliers. It is the view of the Working Group and the research team, ranking a project as a “5” should be extremely rare, perhaps only once in a career. Absent clear direction, project teams can become convinced nearly every job is a, “5”, because when they are in the “heat of battle” it seems that way to them.

B. DATA ANALYSIS

The intent of this research was not to create a predictive model for labor productivity impacts in the Painting Trades, though undoubtedly, some will see our work in that way. In today’s construction environment, new tools linking handheld

⁹ Some like the MCAA labor productivity impact guide document no actual projects, as the basis of their findings.

tablets to Computer Aided Design Drawings (CADD) are becoming more common in the field. As trade contractors collect this data, it is often their purpose to use it to show conditions on the job caused or at least contributed to negative changes in productivity, and therefore ask for additional compensation. In order make these connections, we must show a causal relationship between various impact factors and a reduction in labor productivity. To accomplish this, two statistical methods were applied: a correlation matrix of the variables; and, single linear regressions of the impact factors and the relative productivity for the job.

1. Master Correlation Matrix

The first method applied was to take all of the (15) impact variables and the Relative Productivity for the three trades (finishing, commercial, and industrial painting) and prepare a master correlation matrix for each. The Relative Productivity is the only calculated variable in the data sets. It is calculated by taking the estimated productivity divided by the actual productivity, obtained from the project data collection sheets discussed above (and found in Appendix B).

The second step was to examine how all 15 impact variables are correlated, and most importantly, how they are related to the Relative Productivity variable. The correlations can range from -1.0 to +1.0, negative 1.0 indicating the variables are 100% negatively correlated, and positive 1.0 meaning they are 100% positively correlated. In the correlation matrix shown below, the middle diagonal correlations have been deleted since all are 1.0, and are reflective of the fact these are comparisons of the variable to itself.

Correlations greater than 0.5 have been highlighted in red. As in earlier, similar research [7], The general contractor controlled variables appear to be highly, positively correlated. In the matrix above, the general contractor coordination is highly correlated with seven other impact variables including five of the other general contractor controlled issues: GC's Contract Administration; Congestion; Fragmentation; Overtime; and Rework. Rework is highly, positively correlated with six other impact variables, also general contractor controlled. Again here, Fragmentation and Congestion are highly, positively correlated at 0.69, meaning when fragmentation of the work occurred, congestion occurred at the or near the same level of impact (on the 1 to 5 scale) nearly 70% of the time. Also, general contractor's coordination (or lack of) is highly correlated to impact variables for congestion, fragmentation, overtime, and rework. Rework was determined to be highly correlated to each of those same impact variables, which is probably expected, given when project has problems, they have to be fixed, and that usually involves rework. Finally, it should be noted that all of the impact variables are negatively correlated to project relative productivity. This means when the issues described by these variables are present, labor productivity suffers, and the higher the ranking of the variable (1 to 5) the lower the relative labor productivity. However, not all of these correlations were found to be statistically significant. Variables which described: Construction Documents, Local Workforce, and Building Environment were not found to be statistically significant. (The remaining twelve had an average 2-tailed significance of 0.0045; Bid Issues was found to be significant at 0.033, all others were smaller than 0.001).

TABLE 1 CORRELATION MATRIX FOR COMMERCIAL PAINTING

	PRODUCTIVITY	CONSTR. DOCS	OVERZEALOUS INSP	WEATHER	WORK STOPAGES	ACCESS TO THE WORK	BID ISSUES	LOCAL WORKFORCE	VENDORS/DELIVERIES	GC'S COORDINATION	BUILDING ENVIRON	GC CONTRACT ADMIN	CONGESTION	FRAGMENTATION	OVERTIME	REWORK
PRODUCTIVITY		-0.15	-0.40	-0.26	-0.44	-0.23	-0.18	-0.10	-0.24	-0.25	-0.07	-0.27	-0.28	-0.31	-0.28	-0.29
CONSTR. DOCS	-0.15		0.40	0.49	0.27	0.28	0.37	0.16	0.29	0.21	0.30	0.44	0.12	0.14	0.23	0.25
OVERZEALOUS INSP	-0.40	0.40		0.26	0.47	0.25	0.36	-0.02	0.38	0.51	0.01	0.50	0.36	0.46	0.59	0.48
WEATHER	-0.26	0.49	0.26		0.55	0.39	0.31	0.09	0.19	0.11	0.32	0.32	0.01	0.05	0.20	0.19
WORK STOPAGES	-0.44	0.27	0.47	0.55		0.58	0.30	0.13	0.27	0.37	0.24	0.53	0.31	0.32	0.45	0.39
ACCESS TO THE WORK	-0.23	0.28	0.25	0.39	0.58		0.16	0.23	0.01	0.51	0.41	0.48	0.38	0.39	0.28	0.30
BID ISSUES	-0.18	0.37	0.36	0.31	0.30	0.16		0.29	0.01	0.21	0.06	0.27	0.00	0.05	0.26	0.27
LOCAL WORKFORCE	-0.10	0.16	-0.02	0.09	0.13	0.23	0.29		0.04	-0.06	0.04	-0.01	-0.05	-0.04	-0.03	-0.02
VENDORS/DELIVERIES	-0.24	0.29	0.38	0.19	0.27	0.01	0.01	0.04		-0.14	-0.07	0.07	0.12	0.07	0.01	0.08
GC'S COORDINATION	-0.25	0.21	0.51	0.11	0.37	0.51	0.21	-0.06	-0.14		0.31	0.59	0.51	0.56	0.69	0.63
BUILDING ENVIRON	-0.07	0.30	0.01	0.32	0.24	0.41	0.06	0.04	-0.07	0.31		0.41	0.36	0.32	0.04	0.52
GC CONTRACT ADMIN	-0.27	0.44	0.50	0.32	0.53	0.48	0.27	-0.01	0.07	0.59	0.41		0.50	0.52	0.44	0.62
CONGESTION	-0.28	0.12	0.36	0.01	0.31	0.38	0.00	-0.05	0.12	0.51	0.36	0.50		0.69	0.44	0.55
FRAGMENTATION	-0.31	0.14	0.46	0.05	0.32	0.39	0.05	-0.04	0.07	0.56	0.32	0.52	0.69		0.46	0.53
OVERTIME	-0.28	0.23	0.59	0.20	0.45	0.28	0.26	-0.03	0.01	0.69	0.04	0.44	0.44	0.46		0.57
REWORK	-0.29	0.25	0.48	0.19	0.39	0.30	0.27	-0.02	0.08	0.63	0.52	0.62	0.55	0.53	0.57	

TABLE 2 CORRELATION MATRIX FOR INDUSTRIAL PAINTING

	PRODUCTIVITY	CONSTR. DOCS	OVERZEALOUS INSP	WEATHER	WORK STOPAGES	ACCESS TO THE WORK	BID ISSUES	LOCAL WORKFORCE	VENDORS/DELIVERIES	GC'S COORDINATION	BUILDING ENVIRON	GC CONTRACT ADMIN	CONGESTION	FRAGMENTATION	OVERTIME	REWORK
PRODUCTIVITY		0.07	-0.28	-0.06	-0.05	-0.24	-0.21	-0.11	-0.15	-0.09	0.13	-0.26	-0.20	-0.21	0.12	-0.06
CONSTR. DOCS	0.07		0.14	0.28	0.27	0.00	0.23	-0.25	-0.20	0.66	0.43	0.29	0.55	0.48	0.73	0.46
OVERZEALOUS INSP	-0.28	0.14		0.40	0.20	0.38	0.25	0.30	0.49	0.51	0.07	0.75	0.55	0.40	-0.04	0.16
WEATHER	-0.06	0.28	0.40		0.28	0.26	0.33	0.13	0.37	0.34	0.77	0.39	0.40	0.29	0.18	0.04
WORK STOPAGES	-0.05	0.27	0.20	0.28		0.21	0.23	-0.13	-0.13	0.21	0.20	0.23	0.12	0.38	0.23	-0.14
ACCESS TO THE WORK	-0.24	0.00	0.38	0.26	0.21		0.53	-0.03	0.10	0.23	0.13	0.24	0.51	0.54	-0.23	-0.24
BID ISSUES	-0.21	0.23	0.25	0.33	0.23	0.53		-0.14	0.08	0.46	0.49	0.20	0.50	0.77	0.17	0.25
LOCAL WORKFORCE	-0.11	-0.25	0.30	0.13	-0.13	-0.03	-0.14		0.44	-0.13	-0.17	-0.07	-0.12	-0.23	-0.16	-0.07
VENDORS/DELIVERIES	-0.15	-0.20	0.49	0.37	-0.13	0.10	0.08	0.44		-0.07	0.15	0.23	0.07	-0.18	-0.16	0.03
GC'S COORDINATION	-0.09	0.66	0.51	0.34	0.21	0.23	0.46	-0.13	-0.07		0.36	0.59	0.77	0.63	0.47	0.40
BUILDING ENVIRON	0.13	0.43	0.07	0.77	0.20	0.13	0.49	-0.17	0.15	0.36		0.08	0.24	0.31	0.32	0.03
GC CONTRACT ADMIN	-0.26	0.29	0.75	0.39	0.23	0.24	0.20	-0.07	0.23	0.59	0.08		0.63	0.49	0.21	0.37
CONGESTION	-0.20	0.55	0.55	0.40	0.12	0.51	0.50	-0.12	0.07	0.77	0.24	0.63		0.66	0.43	0.39
FRAGMENTATION	-0.21	0.48	0.40	0.29	0.38	0.54	0.77	-0.23	-0.18	0.63	0.31	0.49	0.66		0.39	0.46
OVERTIME	0.12	0.73	-0.04	0.18	0.23	-0.23	0.17	-0.16	-0.16	0.47	0.32	0.21	0.43	0.39		0.61
REWORK	-0.06	0.46	0.16	0.04	-0.14	-0.24	0.25	-0.07	0.03	0.40	0.03	0.37	0.39	0.46	0.61	

TABLE 3 CORRELATION MATRIX FOR TAPING AND FINISHING

	PRODUCTIVITY	CONSTR. DOCS	OVERZEALOUS INSP	WEATHER	WORK STOPAGES	ACCESS TO THE WORK	BID ISSUES	LOCAL WORKFORCE	VENDORS/DELIVERIES	GC'S COORDINATION	BUILDING ENVIRON	GC CONTRACT ADMIN	CONGESTION	FRAGMENTATION	OVERTIME	REWORK
PRODUCTIVITY		-0.32	-0.34	-0.30	0.01	-0.31	-0.27	-0.14	-0.10	-0.40	-0.28	-0.34	-0.28	-0.40	-0.27	-0.32
CONSTR. DOCS	-0.32		0.58	0.51	0.16	0.21	0.24	0.00	0.14	0.34	0.39	0.51	0.32	0.28	0.48	0.30
OVERZEALOUS INSP	-0.34	0.58		0.60	0.11	0.18	0.40	0.18	0.18	0.44	0.29	0.51	0.37	0.47	0.49	0.39
WEATHER	-0.30	0.51	0.60		0.32	0.25	0.45	0.15	0.03	0.36	0.37	0.42	0.37	0.45	0.36	0.29
WORK STOPAGES	0.01	0.16	0.11	0.32		0.23	0.11	-0.03	0.15	0.11	0.15	0.07	0.09	0.12	0.20	0.09
ACCESS TO THE WORK	-0.31	0.21	0.18	0.25	0.23		0.33	0.12	0.12	0.60	0.39	0.54	0.67	0.55	0.34	0.48
BID ISSUES	-0.27	0.24	0.40	0.45	0.11	0.33		0.42	0.13	0.34	0.13	0.22	0.37	0.55	0.39	0.48
LOCAL WORKFORCE	-0.14	0.00	0.18	0.15	-0.03	0.12	0.42		0.31	0.18	0.13	0.09	0.12	0.38	0.09	0.27
VENDORS/DELIVERIES	-0.10	0.14	0.18	0.03	0.15	0.12	0.13	0.31		0.27	0.15	0.14	0.07	0.09	0.05	0.05
GC'S COORDINATION	-0.40	0.34	0.44	0.36	0.11	0.60	0.34	0.18	0.27		0.40	0.64	0.64	0.69	0.39	0.55
BUILDING ENVIRON	-0.28	0.39	0.29	0.37	0.15	0.39	0.13	0.13	0.15	0.40		0.60	0.62	0.42	0.34	0.40
GC CONTRACT ADMIN	-0.34	0.51	0.51	0.42	0.07	0.54	0.22	0.09	0.14	0.64	0.60		0.68	0.58	0.35	0.47
CONGESTION	-0.28	0.32	0.37	0.37	0.09	0.67	0.37	0.12	0.07	0.64	0.62	0.68		0.64	0.44	0.56
FRAGMENTATION	-0.40	0.28	0.47	0.45	0.12	0.55	0.55	0.38	0.09	0.69	0.42	0.58	0.64		0.47	0.69
OVERTIME	-0.27	0.48	0.49	0.36	0.20	0.34	0.39	0.09	0.05	0.39	0.34	0.35	0.44	0.47		0.56
REWORK	-0.32	0.30	0.39	0.29	0.09	0.48	0.48	0.27	0.05	0.55	0.40	0.47	0.56	0.69	0.56	

Again, we have highlighted in red, those variables which are greater than 0.50. Here again we seen some of the same patterns, again congestion and fragmentation are highly correlated and GC's Coordination is highly correlated to both, as well as GC Contract Administration. The majority, but not

all variables are negatively correlated with relative productivity. But here is the caution, only two of the variables were determined to be statistically significant: Overzealous Inspection (at 0.034) and GC Contract Administration (at 0.049) when compared with Relative Productivity.

The lack of statistically significant relationships between the impact variables and relative productivity, may be the product of fewer industrial project (recall there were only 57 industrial painting jobs compared with 140 commercial painting jobs) or maybe because there are a wider variety of industrial applications. Whereas most commercial painting jobs are painting building interiors, industrial applications range from water towers to bridges to power plants. It should be pointed out however, there were some highly correlated, and statically significant intra-impact variable relationships. For example, Fragmentation and Congestion were positively correlated at positive 0.66, and the two-tailed significance was found to be 2.77×10^{-8} and Overtime was correlated to Construction Documents at 0.77 and found to be significant at 1.66×10^{-10} (though we would be the first to point out, these ultralow calculated significance values are pretty much meaningless, since they are simply less than 0.01, the normal standard).

The final correlation matrix shown above is for the taping and finishing of gypsum drywall. This output is very similar to what was originally found and shown in [10]. The general contractor controlled variables are again shown to be highly correlated and statistically significant. The average of all (including those not greater than 0.50) variables with the general contractor controlled issues, with GC's Coordination is 0.52 and the average level of statistical significance is 2.41×10^{-6} (noting again the same caveat as above regarding these small significance level numbers). All the impact variables are negatively correlated with Relative Productivity, but three of these variables: Work Stoppages, Local Workforce, and Vendors/Deliveries, were not found to be statistically significant (with levels of: 0.89, 0.09, and 0.27 respectively).

We also want to point out one of the fallacies of the "correlation" versus "causation" argument which is revealed above. Note: Construction Documents are highly correlated with Weather! And, the statistical significance level was calculated to be 3.63×10^{-11} . Weather is even more highly correlated with Overzealous Inspection, and found to have a statistical significance level of 5.35×10^{-16} . There may be a plausible explanation for the relationship between bad weather and overzealous inspectors; perhaps when the weather gets bad, the inspectors get mad? It's also possible, since this data came from contractors across the country, in regions of the country where the weather happens to be "good" most of the time, the inspectors are easier on the contractors; whereas, in areas which have "bad" weather, the inspectors are tougher? Nevertheless, there is no sensible argument we can make for the relationship between bad weather and poor quality of construction documents.

The important thing to point out, is this analysis, consistent with the Northwest Wall & Ceiling Bureau (NWCBC) study [10], shows high, statistically significant levels of correlation between variables controlled by the general contractor. It supports the earlier conclusion, when one part of the project starts going bad, such as suffering high levels of congestion, other bad things happen, like high levels of fragmentation,

rework, and overtime. Also, even though many issues occur on a project simultaneously, these high levels of correlation mean most of these variables cannot be combined to justify greater impacts in productivity.

2. Simple Single Linear Regressions

The simple linear regression step was used to find which causes and effects have the strongest relationship to lost productivity and validate that the responses were consistent with the literature on productivity losses. In this step, each of the 15 variables were regressed against the single "Relative Productivity" variable. Conceptually this is the same as plotting all of the projects on an X-Y axis, where the Relative Productivity is plotted on the vertical (Y) axis and the responses (1 to 5) are plotted on the horizontal (X) axis. Regression is then a systematic process for drawing the best fits the data. If the line is "tilted" downwards, left to right (1 to 5 on the X-axis) reflecting a negative slope and indicating an increase in the X-axis variable, relates to a decrease in Relative Productivity.

TABLE 4 SIMPLE LINEAR REGRESSION FOR COMMERCIAL PAINTING

SINGLE LINEAR REGRESSIONS	
	RELATIVE PRODUCTIVITY
CONSTR. DOCS	NOT STA SIG
OVERZEALOUS INSP	-0.10
WEATHER	-0.08
WORK STOPAGES	-0.14
ACCESS TO THE WORK	-0.08
BID ISSUES	-0.06
LOCAL WORKFORCE	NOT STA SIG
VENDORS/DELIVERIES	-0.10
GC'S COORDINATION	-0.07
BUILDING ENVIRON	NOT STA SIG
GC CONTRACT ADMIN	-0.08
CONGESTION	-0.07
FRAGMENTATION	-0.08
OVERTIME	-0.10
REWORK	-0.08

As is known from the discussion of the Correlation Matrices above, three of the Commercial Painting impact variables were not found to be statistically significantly related to Relative Productivity of trade labor. However, twelve do have a statistically significant relationship with Relative Productivity. The lowest impact is -0.06 (Bid Issues) and the highest -0.14 (Work Stoppages).

There are probably a couple of reasons why Construction Documents and Local Workforce were not found to be statistically significant. First, the painting contractor is usually just painting walls, and relies on the plans merely to figure out how many square feet of wall are shown on the plans. The contract specifications tell the contractor the quality or type of paint and possibly the number of coats applied, and expected finish. Unless there is some dramatic change in the building

design, at the end of the job, they are probably going to have painted the same amount of wall area as anticipated at bid time. As far as the local workforce is concerned, it is likely most of the projects in the database are reasonable “hometown” projects. Local Workforce problems tend to occur when a contractor travels outside its normal sphere of operations, and that is probably unlikely in this case. Secondly, the majority of the respondents are union contractors who hire out of the union hall. The union is in place in part to guarantee its members have some level of training and can perform at an expected level of productivity. Finally, perhaps painting a wall is not so complex, compared with mechanical and electrical operations which vary widely from building design to building design. Painting is pretty much the same from building to building.

TABLE 5 SIMPLE LINEAR REGRESSIONS FOR INDUSTRIAL PAINTING

SINGLE LINEAR REGRESSIONS	
	RELATIVE PRODUCTIVITY
CONSTR. DOCS	NOT STA SIG
OVERZEALOUS INSP	-0.10
WEATHER	NOT STA SIG
WORK STOPAGES	NOT STA SIG
ACCESS TO THE WORK	NOT STA SIG
BID ISSUES	NOT STA SIG
LOCAL WORKFORCE	NOT STA SIG
VENDORS/DELIVERIES	NOT STA SIG
GC'S COORDINATION	NOT STA SIG
BUILDING ENVIRON	NOT STA SIG
GC CONTRACT ADMIN	-0.09
CONGESTION	NOT STA SIG
FRAGMENTATION	NOT STA SIG
OVERTIME	NOT STA SIG
REWORK	NOT STA SIG

As noted above, only two of the impact variables had a statistically significant relationship to Relative Productivity: Overzealous Inspections and GC Contract Administration. The smaller number of projects in Industrial Painting, 57 compared with 140 commercial jobs, and the wider variety of types of industrial jobs may have contributed to the lack of statistically significant relationships with Relative Productivity. Though, since many industrial jobs are outside in the weather, it is curious the Weather variable was not found to be statistically significant. (It could be when the weather turns “bad” the industrial painters simply don’t work, and the contractor receives a time extension to complete its work, consistent with the number of days the weather shut the job down.) It will take more than a statistical analysis of the data to shed light on these issues. What is not surprising is Overzealous Inspections have a relatively strong (-0.10) impact and a statistically significant relationship with Relative Productivity. Though, if there are overzealous inspections, resulting in rejection of the work, one would expect to see Rework as an issue as well, and likewise work fragmentation.

TABLE 6 SIMPLE LINEAR REGRESSIONS FOR DRYWALL FINISHING

SINGLE LINEAR REGRESSIONS	
	RELATIVE PRODUCTIVITY
CONSTR. DOCS	-0.11
OVERZEALOUS INSP	-0.09
WEATHER	-0.09
WORK STOPAGES	NOT STA SIG
ACCESS TO THE WORK	-0.10
BID ISSUES	-0.08
LOCAL WORKFORCE	NOT STA SIG
VENDORS/DELIVERIES	NOT STA SIG
GC'S COORDINATION	-0.10
BUILDING ENVIRON	-0.09
GC CONTRACT ADMIN	-0.08
CONGESTION	-0.06
FRAGMENTATION	-0.08
OVERTIME	-0.11
REWORK	-0.09

Three of the impact variables did not have a statistically significant relationship to Relative Productivity, the other twelve ranged from -0.06 (Congestion) to -0.11 (Construction Documents and Overtime), the average of the twelve is -0.09. It is somewhat surprising Congestion and Fragmentation (-0.06 and -0.08) were not bigger impacts, however in [7], the combined variable representing Work Fragmentation found the impact to be -0.056, and the combined Congestion variable -0.085, one lower, the other higher.

We note none of the trades found a statistically significant relationship between Local Workforce and Relative Productivity. This could be explained, as mentioned above, the respondents may not have had many projects away from their “hometowns.” Vendors/Deliveries was not found to be statistically significant for the Finishing trade, where it was significant to the commercial painters. Painters may require more deliveries to maintain a constant flow of work compared with finishers, who don’t usually cover the entire wall with joint compound. In general, we note the impacts are greater in this study compared with the earlier NWCB work, we attribute this to a better designed survey instrument, which defined the levels of impact, 1 to 5, instead of leaving it up to the contractor’s.

IV. RESULTS, CONCLUSIONS, AND FUTURE WORK

This work was the first major study of the commercial and industrial painting and the second of drywall finishers. 140 commercial, and 57 industrial painting projects were studied, along with 147 gypsum drywall tape and finishing jobs. The analyses showed negative relationships between the impact variables and the production rates achieved on the projects. With the exception of the industrial painting jobs, the majority (12 of 15 in both cases) of the impact variables demonstrated a statistically significant (negative) relationship with productivity. The outstanding question is why the industrial

painting projects differed so greatly from the other two sets of data?

The fundamental conclusion of this research, are that factors outside the trade contractor's control, including but not limited to: overzealous inspection, access to the work, the general contractor's coordination and administration of the work, workplace congestion, work fragmentation, overtime, and rework, have been shown to have a statically significant correlation to a negative impact on the contractor's labor productivity. Consequently, the trade contractor may be entitled to seek a remedy of compensation or additional time from the general contractor or owner on the project.

It is important to point out, in today's modern construction environment, the ability to collect production data using new technology devices, such as electronic devices linked to CADD files, allows the contractor to analyze and evaluate labor productivity impacts contemporaneously. Furthermore, when new technologies exist which allow the contractor to capture these production data and the contractor fails to do so, courts and arbitrators tend to hold it against them. Nevertheless, even having the data is unhelpful if there is no way to connect the productivity impacts with the environmental factors which occurred on the job. The research makes that linkage.

This research builds on the already substantial body of labor productivity impact research in the construction trades, and for the first time, adds commercial and industrial painting to trades previously studied. Drywall finishing had been studied previously by some of the same researchers here, but in this study, we feel better context in the data collection has yielded better responses from the trade contractors.

There is certainly future work to be done in this field, including more work using the data presented in this paper. For example, the respondents provided information on schedule performance and the amount of change orders, which were not addressed in this paper. The type of commercial building also was not addressed in this study and it would be interesting to know if there are interactions between productivity impacts and project type. Size is another issue not explicitly studied here; do smaller projects yield significantly higher impacts compared to larger jobs? Intuitively, when a contractor faces a large project, there are greater opportunities to recover from short term exogenous conditions, whereas a small job may not. Also, this research did not ask respondents to provide a narrative of the problems or conditions experienced in the execution of the work. This could provide further insight to the issues studied here.

Improving the responses from the industrial sector is also a challenge for future research. While 57 projects is not as great a number of responses as received for commercial painting and drywall finishing, it is certainly a much larger number of projects compared with many other studies in the construction industry. (Some of which looked at fewer than a dozen jobs.) So the fundamental question remains, why weren't the responses in the industrial sector statistically significant for all

but two factors? Is it because industrial painting includes such diverse activities as painting a bridge over water and painting a water tank on land? At this point we simply do not know.

Finally, while many studies on labor productivity in the construction trades have been performed, there are many trades which have not been studied at all and others where more work could certainly be done.

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APPENDIX A

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APPENDIX B

A Contractor's Guide to the Labor Productivity Survey

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**FINISHING
CONTRACTORS
ASSOCIATION**

Impacts to Labor Productivity *in the drywall, floor covering, painting, and glazing finishing industries.*

What if you had the **data to support a claim** without actually having to go to the expense of engaging consultants or attorneys?

That's the goal of this FCA Labor Productivity Survey, which grew out of a concern that we all face common factors that impact job completion, no matter where the location or the type of work. Inside a building we're all in the same boat.

Dr. Gerald H. Williams, whose Portland, Oregon firm, Construction Research, Inc., will compile the results of hundreds of jobs performed by FCA contractors, says this study provides the key "entitlement" element for negotiating changes and that there are factors common to all trades.

That idea was supported by Dr. Williams' most recent study for the Northwest Wall & Ceiling Bureau.



Industry subcontractors already see the value of the study, said Tina Freeman, spokeswoman for the Northwest Wall and Ceiling Bureau. "It will make it a lot easier for them to recover costs for factors outside of their control, such as if the general contractor wants them to accelerate the schedule, or if other trades are working at the same time," she said.

Other trades have documents they can use to back up claims of lost productivity, but drywall contractors have been behind in that area," Freeman said. "They know that those factors are affecting productivity but, until now, they haven't had a specific document to point to."



The major finding of the NWCB study was that **all of the major impact variables negatively correlated with labor productivity** (meaning the more severe the impact, as recorded on a scale of one to five, the greater the impact to labor productivity).



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Complete this survey **three times** for **three different projects**, one **good** project, one **bad**, and one **average**.

Respondent

Company

Project Name

Contact information is for the purpose of questions or clarifications on this survey. Your information will not be shared with other FCA members.

It is not necessary to include the legal name of the project; however, a name (even if it is a made-up identifier) is necessary for reference purposes during the data analysis phase.

1a. How many weeks were your crews **supposed** to be on site, according to the schedule, before the project began?

estimated weeks

1b. How long were your crews **actually** on site?

actual weeks

2. Dollar value of contract:

Original \$

Change Order Total \$

Final Contract Total \$

3. Type of Work

Commercial Painting & Decorating
Drywall
Glass/Glazing
Floor Covering
Industrial Painting

4. Type of Project

school (K-12)
institutional
power plant
water tower
residential

Other

5. Check all that apply:

- ☐ Fast Track
☐ Private Sector
☐ Public Sector

FCA: Labor Productivity Survey

Sample Responses to Question 6

Question 6 deals with a **SPECIFIC STEP** or **TYPE OF WORK** in the process. There are 5 blanks here, but you may need more or less, depending upon the work done for your project. Each blank represents the specific **MATERIAL APPLICATION** or **TASK PERFORMED**.

It will be different for each of the finishing trades, and it will be unique to your project. The examples (one for each trade) show you how to complete one type of work for each trade.

A= Flooring

B= Glazing

C= Commercial Painting

D= Tape & Finishing

E= Industrial Painting

A. Type of Work Carpet	Estimated Quantity 200	Units Square Feet	
	Estimated Productivity 50 SY/Man hour	Actual Productivity 40	Units Square Yards/MH
B. Type of Work Glazing	Estimated Quantity 20,000	Units Square Feet	
	Estimated Productivity 40 SF/Man Day	Actual Productivity 50	Units Square Feet/MD
C. Type of Work Painting	Estimated Quantity 600	Units Square Feet	Wall = Length x Height
	Estimated Productivity 300	Actual Productivity 2 hours	Units Gallons
D. Type of Work Drywall Finishing: Level 4 walls through ceiling.	Estimated Quantity 20,000	Units Square Feet	
	Estimated Productivity 1,000 sf/man day	Actual Productivity 783	Units Square Feet/Man Day
E. Type of Work Industrial Painting	Estimated Quantity 400,000	Units Square Feet	
	Estimated Productivity 500 Square Feet/Man Hour	Actual Productivity 400	Units Square Feet/MH

6. Type and quantity of work, and estimated and actual productivities for multiple types of work: Fill in the boxes of each work type or process you did within the project.

A. Type of Work			
Estimated Quantity		Units	
Productivity			
Estimated		Actual	

B. Type of Work			
Estimated Quantity		Units	
Productivity			
Estimated		Actual	

C. Type of Work			
Estimated Quantity		Units	
Productivity			
Estimated		Actual	

D. Type of Work			
Estimated Quantity		Units	
Productivity			
Estimated		Actual	

E. Type of Work			
Estimated Quantity		Units	
Productivity			
Estimated		Actual	

F. Type of Work			
Estimated Quantity		Units	
Productivity			
Estimated		Actual	

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Directions:

Rank questions 1-5. Rankings are based on a scale of 1 to 5, where 1 is the least severe and 5 is the most severe. Rankings are explained as they pertain to each question.

Owner Related Issues

1.	Construction Documents: Rate the overall quality of the plans and specifications relative to what you think is required to execute your scope of work.		<<
	1. <i>Standard of the Industry. Requires minimal requests for Information from designer.</i> 2. <i>Could not be built without some direction from designer.</i> 3. <i>Required weekly direction to execute your scope of work.</i> 4. <i>Almost daily direction.</i> 5. <i>Plans almost useless. Final project different from original design documents.</i>		R A N K
2.	Overzealous and Unreasonable Construction Administration: Rate the owner's construction administration on the project, which includes: overzealous inspections, unreasonable safety requirements, or any other interference from the owner's personnel that distracts or impacts worker production.		<<
	1. <i>Little or no interference</i> 2. <i>Higher than normal interference</i> 3. <i>Strict adherence to requirements such as measuring distances exactly and rejecting an abnormal amount of work for minor deviations in specifications.</i> 4. <i>Daily run-ins with inspectors and other owner-controlled administrators</i> 5. <i>Owner-controlled interferences make job so unbearable that you threaten to quit.</i>		R A N K
3.	Weather: Rate the weather during the project, relative to what the estimator expected when the job was bid, to the extent that weather affected worker productivity on the job.		<<
	1. <i>No unaccounted for (in the estimate) or unusual weather events that affect labor productivity</i> 2. <i>Minor change in weather-related environment (for example, push summer work to fall)</i> 3. <i>Major change in weather-related environment</i> 4. <i>Multiple work stoppages due to weather</i> 5. <i>Job halted for multiple days (Hurricane Katrina event)</i>		R A N K
4.	Work Stoppages, Acts of God, or the Public Enemy: Rate whether work stoppages on the job affected your labor production.		<<
	1. <i>No stoppages</i> 2. <i>Minor or occasional work stoppages, (e.g. strikes that didn't affect your workers but held them up)</i> 3. <i>Minor work stoppages of your crews</i> 4. <i>Major work stoppages</i> 5. <i>Major events (e.g. labor dispute, strike for months that shuts down job)</i>		R A N K
5.	Access to Jobsite or the Work Issues: Access to the jobsite or the work covers a broad number of issues, such as the adequacy and location of parking relative to the jobsite, adequacy of vertical transportation in multi-story buildings, location of and adequacy of the restrooms, eating areas, and other support facilities.		<<
	1. <i>No unaccounted for access problems</i> 2. <i>Minor access issues</i> 3. <i>Major access issues</i> 4. <i>Major, unaccounted for access issues, requiring contractor to provide alternative access</i> 5. <i>Access frequently blocked</i>		R A N K

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Directions:

Rank questions 6-8. Rankings are based on a scale of 1 to 5, where 1 is the least severe and 5 is the most severe. Rankings are explained as they pertain to each question.

Subcontractor Related Issues - Rate 1-5

6.	Bid or Estimate: Were there issues with the contractor's bid, such as missing scope or overly optimistic productivities used in the estimate?		<<
1.	<i>No bid issues</i>		R A N K
2.	<i>Minor holes in the bid</i>		
3.	<i>Major errors in bid, such as leaving out all supervision or overhead</i>		
4.	<i>Major portions of work were missing or forecast productivities were absurdly high</i>		
5.	<i>Estimator never looked at plans, used standard productivities, greater than 50%</i>		
7.	Unaccounted for Local Workforce: This factor is specific to the local workforce, meaning that local workforce productivities vary significantly from the productivities anticipated in the bid.		<<
1.	<i>No workforce issues</i>		R A N K
2.	<i>Worse than expected (less than 25% reduction in productivity)</i>		
3.	<i>High impact (25-50% reduction in productivity)</i>		
4.	<i>Greater than 50– 75% reduction in productivity</i>		
5.	<i>Greater than 75%</i>		
8.	Problems with Vendors: This factor describes problems you experienced on the job that were generated by your own vendors or suppliers.		<<
1.	<i>No delivery problems</i>		R A N K
2.	<i>Minor delays in deliveries, causing occasional resource reallocation</i>		
3.	<i>Multiple delays in supply chain. Deliveries unreliable</i>		
4.	<i>Severe Delivery Problems forcing you to seek suppliers from elsewhere</i>		
5.	<i>Cannot obtain supplies causing a default or other major change</i>		

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Directions:

Rank questions 9-13. Rankings are based on a scale of 1 to 5, where 1 is the least severe and 5 is the most severe. Rankings are explained as they pertain to each question.

General Contractor Issues - Rate 1-5

9.	Coordination: This factor describes the coordination problems experienced by you that are the responsibility of the general contractor.		<<
	1. No coordination problems 2. Minimal — materials a day or week late 3. Out of sequence work installed and had to be torn out 4. Coordination non-existent, job becomes "First come. First installed" 5. Affects nearly 100% of job		R A N K
10.	Building Environment: The factor describes the actual working environment relative to what the estimator expected; it includes, but is not limited to heat and humidity.		<<
	1. Environmental issues were no different from plan 2. Slight or moderate differences from plan 3. Major difference from expectations 4. Severe problems with heat and water 5. Work stopped due to the interior environment		R A N K
11.	GC Contract Administration: This factor characterizes the general contractor's (GC) contract administration during the course of the project.		<<
	1. GC met or beat all contractual timeliness. Schedules routinely updated and followed. Safety programs achieved goals and didn't interfere with productivity 2. Minimal delays 3. All review and approval processes delayed. GC in "reactive" mode 4. Review and approval processes unpredictable 5. No visible GC contract administration process resulting in poor workmanship and frequent jobsite injuries		R A N K
12.	Congestion: This factor describes the relative amount of congestion on the project, whether the congestion is caused by "crowding" or "acceleration" of your own workforce, "stacking of trades", or crowding due to materials and equipment in your workspace that your crews have to work around.		<<
	1. Never a problem 2. Moderate, but not highly unusual 3. Highly congested 4. Extremely congested 5. Blocking access to your work, forced to move materials out of way		R A N K
13.	Fragmentation: This factor describes the degree to which the project was fragmented, or broken up, instead of being executed in a fluid manner from start to finish of a given activity.		<<
	1. All work activities executed as planned 2. Moderately fragmented, 1-2 remobilizations 3. More than half performed out of sequence, 5-10 remobilizations 4. None of activities performed as planned, 10 or more remobilizations 5. Every day is a mystery. No continuity or flow of work		R A N K

FCA: Labor Productivity Survey

Directions:

Rank questions 14-15. Rankings are based on a scale of 1 to 5, where 1 is the least severe and 5 is the most severe. Rankings are explained as they pertain to each question.

Continued: General Contractor Issues - Rate 1-5

14. Overtime and Added Shift : This factor describes the extent to which overtime and adding shifts were required to complete the work in a timely fashion, and that were NOT accounted for in the project estimate or bid.		<<
1. No unaccounted for overtime or shifts 2. Minimal overtime, perhaps 1-2 weeks in 6 month project 3. Overtime & shifts added for 33% or more of project 4. 50% of project worked as overtime or extra shifts 5. Unaccounted for overtime and added shifts required		R A N K
15. Building Environment: The factor describes the actual working environment relative to what the estimator expected; it includes, but is not limited to heat and humidity.		<<
1. Rework was NOT extraordinary, but rather was normal, and accounted for in the bid 2. Touch-up or rework was required on a small portion of the work, less than 10% 3. Up to half of the work installed had to be reworked or touched up 4. More than half and up to three quarters of the work had to be touched up or reworked 5. Between 75% and 100% of the job required touch-up and/or rework		R A N K

Questions 16-18 should be answered 'yes' or 'no'.

Schedules

16. Were three-week coordination schedules distributed?	(yes or no)	
17. Was the overall project schedule distributed?	(yes or no)	
18. Were schedule updates distributed?	(yes or no)	

When finished, please save this form, click submit, then choose to 'Reset Form,' and complete the survey two more times, for two other projects, saving each survey as a different file before submitting.

Submit by E-mail

Reset Form

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